

StructureScan™ Mini HR

Quick Start Guide

A Fast Check List For Field Operation



StructureScan Mini HR

Diagram: Top View

Diagram: Bottom View

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GSSI ASSUMES NO LIABILITY FOR ANY DIRECT, INDIRECT, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES OR INJURIES CAUSED BY PROPER OR IMPROPER OPERATION OF ITS EQUIPMENT, WHETHER OR NOT DEFECTIVE.

Before returning any equipment to GSSI, a Return Material Authorization (RMA) number must be obtained. Please call the GSSI Customer Service Manager who will assign an RMA number. Be sure to have the serial number of the unit available.

Regulatory Information

Please see the GSSI Manual CD or our website, www.geophysical.com/regulatoryinformation.htm, for current information and FCC Registration Form, including:

- FCC Notice for U.S. Customer
- Canadian Requirements for RSS-220
- Declaration of CE Conformance

Laser Cautions and Warnings

Please see Appendix D for important laser cautions and warnings.

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Informations Réglementaires

L'utilisation des antennes GSSI est régie par différents organismes de réglementation à travers le monde. Certains modèles d'antenne spécifiques doivent être certifiés pour un fonctionnement légal dans votre pays. Merci de lire et comprendre les passages suivants de réglementation qui s'appliquent à votre antenne. Une liste des antennes certifiées par région peut être trouvée sur [www.geophysical.com / regulatoryinformation.htm](http://www.geophysical.com/regulatoryinformation.htm).

Avis

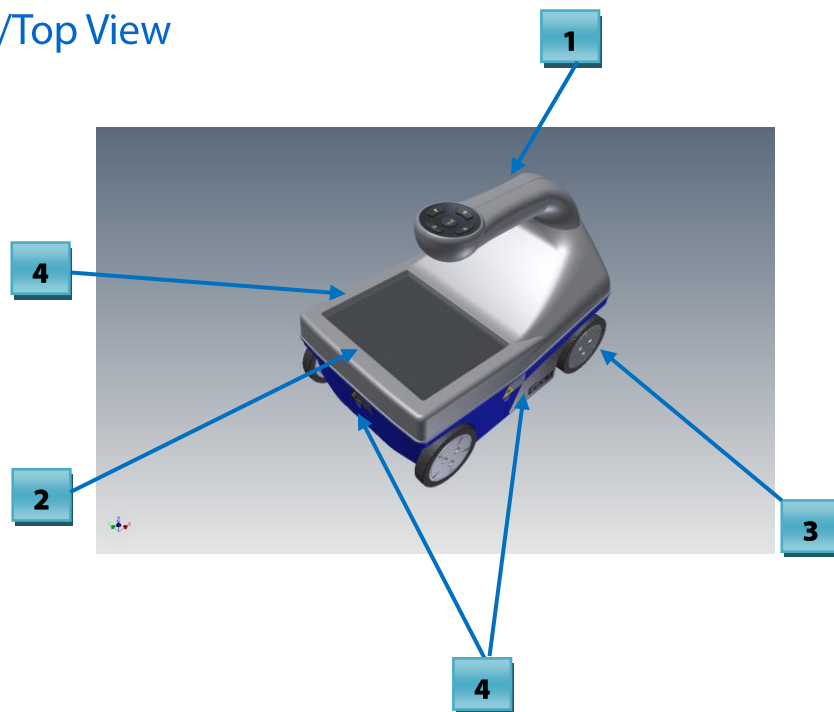
La mise en œuvre est soumise aux deux conditions suivantes: (1) cet appareil ne doit pas provoquer d'interférences et (2) cet appareil doit accepter toute interférence, y compris les interférences qui peuvent causer un fonctionnement non désiré de l'appareil.

Ce dispositif radar à pénétration du sol ne doit être utilisé qu'en contact avec le sol ou à au plus 1 m du sol.

Ce dispositif radar à pénétration du sol ne doit être utilisé que par des organismes d'application de la loi, des établissements de recherche scientifique, des sociétés minières commerciales, des entreprises de construction, et des organismes d'intervention d'urgence ou de lutte contre les incendies

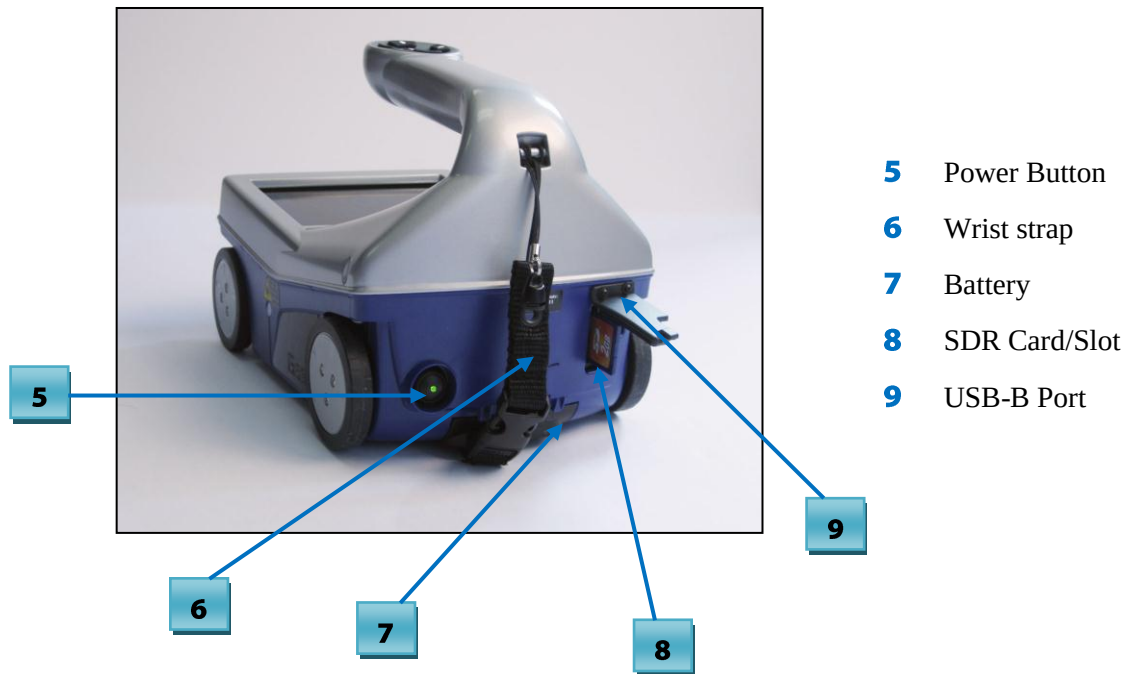
StructureScan Mini HR

Front/Top View



- 1** Ergonomic handle and controls
- 2** Easy to use operator interface with color display screen
- 3** Survey wheel encoder
- 4** Guiding laser for locating

Back View



Section 1: Getting Started

1 Preparing the StructureScan Mini HR.

- a)** Remove the protective plastic cover from the housing and the handle.
- b)** Remove the blue tape from the wheels.
- c)** Attach the wrist strap to the back of the Mini HR. The wrist strap should go around your wrist to prevent the Mini HR from being dropped.



- d)** Attach the lanyard to the back of the Mini HR. The battery lanyard will attach to this lanyard to prevent the battery from falling out of the Mini HR.



- e)** Insert the battery into the battery slot in back of the Mini HR. Lift the tab upward and “snap” the tab into place.



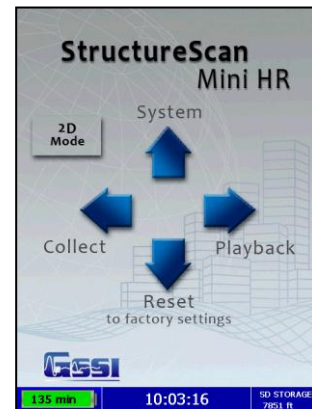
- f)** Attach the battery lanyard to the lanyard on the Mini HR.

2 Powering up the StructureScan Mini HR.

- a)** Press the power button. A green light will illuminate on the power button.



- b)** After a few seconds, the system will boot up and the splash screen (Main Menu) is displayed.
- c)** The Green/Gray bar that is displayed on the lower left hand corner is a battery life indicator. This will display how much battery time is remaining.
- d)** The current time (as stored in the Mini HR) is displayed at the bottom of the screen.
- e)** How much storage you have left (the distance you have left to scan) is displayed on the lower right hand corner.
- This Ground Penetrating Radar Device shall be operated only when in contact with or within 1 m of the ground.
 - This Ground Penetrating Radar Device shall be operated only by law enforcement agencies, scientific research institutes, commercial mining companies, construction companies, and emergency rescue or firefighting organizations.
- 3** To power off the Mini HR, simply press and hold the power button for approximately 2 seconds. Then you may remove the battery by pressing down on the tab of the battery and pulling down.



- 4** Note the buttons on the handle of the Mini HR. They will be referred as Left Arrow ←, Right Arrow →, Up Arrow ↑, Down Arrow ↓, and the Enter ↵ buttons relative to the screen.
- 5** Depending upon the orientation of the screen on the Mini HR (Left or Right depending upon which hand you use), the Left, Right, Up, and Down buttons are switched. However, you will still be pressing ←, →, ↑, and ↓ buttons, regardless of the orientation of the screen.

The Enter button (↵) is located in the center of the keypad Button.

- At any time and unless otherwise noted, you may press the Arrow button closest to the rear of the Mini HR to return to the previous screen or menu. This will be known as the “Back” button.



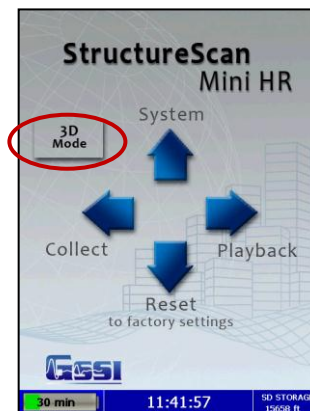
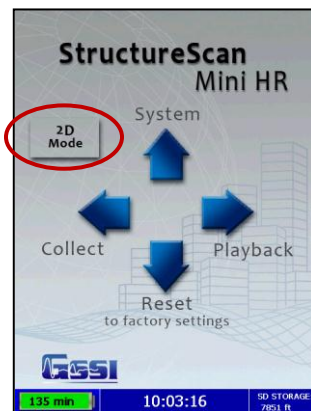
Unless otherwise noted, this button, **regardless of orientation**, will return you to the previous screen or menu.



Section 2: Configuring the Structure Scan Mini HR for 2D or Optional 3D Operation

Before collecting any data, configure your Mini HR for 2D or 3D* data collection by pressing ↵.

Note: See Pages 29-44 on the details of 3D data collection and processing with the Mini HR.



***Note:** The 3D version of the Mini HR is an optional installation. Consult with you GSSI Sales Representative for purchase and installation details.

System Menu Options

- 1 From the Main Menu, select System by pressing **↑**.
- 2 Use **↑** and **↓** buttons to highlight the appropriate options under the System Menu.

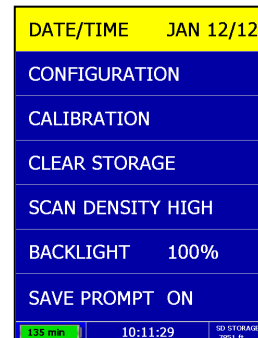
Date/Time: Configuring Date and Time

- 1 Highlight Date/Time and then press **↵**.
- 2 Highlight any of the options you wish to change and then press **↵**.
The following options will appear at the bottom of the screen:

- **Change Value** – Use **↑** and **↓** to change the selected value.
- **Set Value** – Press **↵** to accept your change.
- **Quit** – Press **→** to quit WITHOUT saving the changes.



- 3 Press **→** to return to the previous menu.



2D Mode Configuration: Configuring Orientation, Language, Units, Laser, and Sound

- 1 Highlight Configuration and then press **↩**.
- 2 Press **↑ ↓** to select the appropriate options:
 - **Orientation** – Press **↩** to toggle between Right or Left. This will flip the screen to allow for either right or left handed operation.
 - **Language** – Press **↩** to select your language preference. Language Options are English, Chinese, French and Spanish.
 - **Units** – Press **↩** to toggle between English or Metric.
 - **Laser** – Press **↩** to toggle laser ON or OFF. If configured ON, the laser will light up in the Collect Menu only.
 - **Note:** The laser will blink whether configured ON or OFF when battery life is low (10%).
 - **Caution:** Do not stare into the laser beam. Staring into the laser beam may result in permanent eye damage.
 - **Sound** – Press **↩** to toggle between High, Medium, Low, or OFF.
 - **Version(s)** – Press **↩** to display the version number of the software. Please visit GSSI's web site for updates to the firmware
- 3 Press **→** to return to the previous menu.

ORIENTATION	RIGHT
LANGUAGE	ENGLISH
UNITS	ENGLISH
LASER	ON
SOUND	MEDIUM
VERSION(s)	1.1.5.0mn
OPTIONS	
135 min	10:42:17
SD STORAGE: 30992 ft	

3D Mode Configuration: Configuring Orientation, Language, Units, Laser, and Sound

- 1 Highlight Configuration and then press $\leftarrow$.
- 2 Press $\uparrow$ $\downarrow$ to select the appropriate options:
 - **Orientation** – Press $\leftarrow$ to toggle between Right or Left. This will flip the screen to allow for either right or left handed operation.
 - **Language** – Press $\leftarrow$ to select your language preference. Language options are English, Chinese, French and Spanish.
 - **Units** – Press $\leftarrow$ to toggle between English or Metric.
 - **Laser** – Press $\leftarrow$ to toggle laser ON or OFF. If configured ON, the laser will light up in the Collect Menu only.

Note: The laser will blink whether configured ON or OFF when battery life is low (10%).

 - **Caution:** Do not stare into the laser beam. Staring into the laser beam may result in permanent eye damage.
 - **Sound** – Press $\leftarrow$ to toggle between High, Medium, Low, or OFF.
 - **Version(s)** – Press $\leftarrow$ to display the version number of the software. Please visit GSSI's web site for updates to the firmware.
 - **3D Options** – Press $\leftarrow$ to display the 3D Options menu.
 - **Scan Density** – Press $\leftarrow$ to toggle between Normal and Low.
 - **Mig (ration) Type** – Press $\leftarrow$ to toggle between Half the samples and Max (all the samples).
- 3 Press $\rightarrow$ to return to the previous menu.

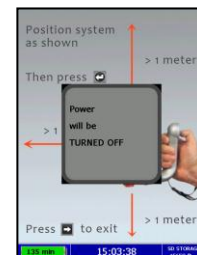
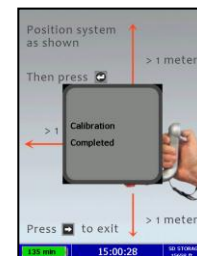
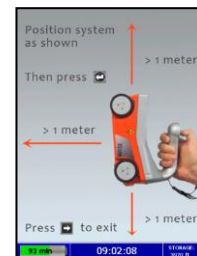
ORIENTATION	RIGHT
LANGUAGE	ENGLISH
UNITS	ENGLISH
LASER	ON
SOUND	MEDIUM
VERSION(s)	1.1.5.0mn
OPTIONS	
135 min	10:42:17
SD STORAGE: 30992 ft	

SCAN DENSITY	LOW
MIG TYPE	HALF
135 min	10:44:13
SD STORAGE: 30992 ft	

SCAN DENSITY	NORMAL
MIG TYPE	MAX
135 min	10:44:13
SD STORAGE: 10440 ft	

Calibration: Auto Calibrations for Surface pulse location and Gain

- 1 Highlight Calibration and press .
- 2 Hold the Mini HR upside down about 1 meter (3 ft) from any surface.
- 3 Press  the Auto-Calibration will take several seconds.
- 4 When the Calibration is finished a Calibration Completed message will be displayed on the screen.
- 5 Press  the Auto-Calibration will take several seconds.
- 6 When the calibration is finished press  and a power down message will appear and the MiniSir will turn OFF. The User must turn the power ON to continue.



Clear Storage: Deleting Saved Data

A project is a folder which holds individual radar files. The entire memory may be deleted, specific projects may be deleted or specific files may be deleted.

- 1 Highlight Clear Storage and press $\leftarrow$.
- 2 At the bottom of the screen, a new set of options is displayed.

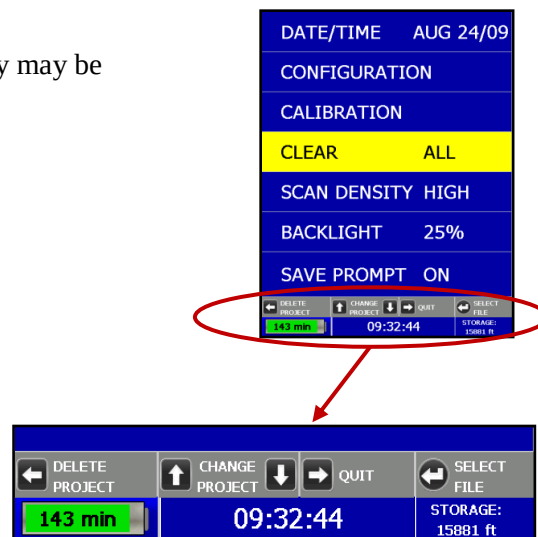
Deleting All Projects, Specific Projects or Specific Files

All Projects: If you wish to delete ALL of your projects and all the files associated with each project:

- 1 Press $\leftarrow$ when Clear All is displayed.
- 2 Press $\rightarrow$ to quit.

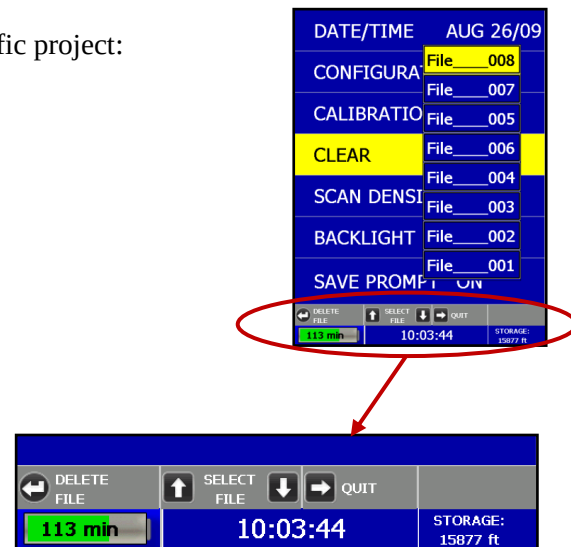
Specific Projects: If you wish to delete a specific project or specific files within a project:

- 1 Press $\uparrow$ $\downarrow$ to select the specific project.
- 2 Press $\leftarrow$ to delete that entire project (and all the files associated with that project).
- 3 Press $\uparrow$ to confirm, $\downarrow$ to cancel.
- 4 At any time, you may press $\rightarrow$ to quit.



Specific Files in a Specific Projects: Deleting specific files within a specific project:

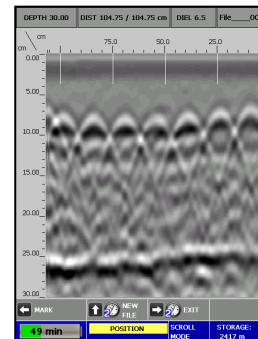
- 1 Press $\uparrow$ $\downarrow$ to select a specific project.
- 2 Press $\leftarrow$ to select that project.
- 3 Press $\uparrow$ $\downarrow$ to select a specific file.
- 4 Press $\leftarrow$ button to delete the selected file.
- 5 Press $\uparrow$ to confirm, $\downarrow$ to cancel.
- 6 At any time, you may press $\rightarrow$ to quit.



Scan Density: Selecting the Horizontal Scan Density

- 1 Highlight Scan Density.
- 2 Press  to toggle between Normal and High.
 - **Normal** – The Mini HR will scan at 4 scans/cm (approximately 120 scans/ft). Use this is when you are looking for normal size objects.
 - **High** – The Mini HR will scan at 8 scans/cm (approximately 240 scans/ft). Use this setting when you are looking for very small objects.

Note: The San Density option is disabled in 3D data collection mode.

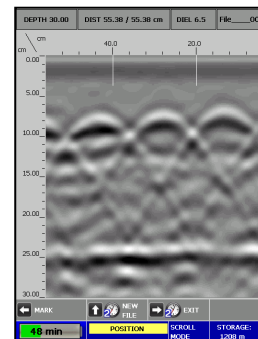


Backlight: Setting the Brightness of the Mini HR Backlight

- 1 Highlight Backlight.
- 2 Press  to toggle between 25%, 50%, 75%, or 100%.

Save Prompt: Have the System Prompt You to Save a File or to Save a File Automatically

- 1 Highlight the Save Prompt.
- 2 Press  to toggle between ON or OFF.
 - **ON** – The system will prompt you to save a file after you choose to save a file.
 - **OFF** – The system will save the file automatically after you choose to save a file.



Section 3: Collecting 2D Data

After the configuration is complete, you are now ready to collect data.

Important Note: To assure consistency of StructureScan Mini HR's naming of the files collected and saved, it is highly recommended that the SD card be inserted prior to any data collection.

Collect Menu Options

- 1 From the Main Menu, select Collect by pressing **←**.
- 2 Use **↑** **↓** buttons to highlight the appropriate options under the Collect Menu.

Before you begin to collect data you may want to configure the Mini HR for a particular scan/project.

START COLLECT	
PROJECT	Proj_1
DEPTH	20 cm
DIELECTRIC	8.0
AUTOTARGET	ON
DISPLAY	B
COLOR	
 14.9 mmHg	15:44:40
STORAGE: 4832 MB	

Project: Saving Files to a Specific Project

- 1 Highlight Project and press **↵**.
- 2 Highlight a Project number and press **↵**. All files that you save during data collection will be saved under this project. (Note: you may want to take field notes to document which files were saved in which project).

Depth: Selecting the Depth You wish to Scan

- 1 Highlight Depth.
- 2 Press **↵** to toggle between 4, 8, 12, and 16 inches (10, 20, 30, and 40 cm).

Dielectric: Selecting the Approximate Dielectric of the Area You are Scanning

1 Highlight Dielectric and press $\leftarrow$ to change the Dielectric value.

- Press $\leftarrow$ to change the number after the decimal point.
- Press $\uparrow \downarrow$ to change the number before the decimal point.
- Press $\leftarrow$ to set your new value.
- Press $\rightarrow$ to quit WITHOUT saving the new value.

Having the correct value of ϵ (Dielectric Constant) will improve your depth accuracy.

A higher value is necessary for young concrete or concrete that is under water.

- Less than 2 months/outside in wet environment, use 9+
- Younger than 12 months/outside, use 7-8
- Older than 12 months/indoor/dry, use 5-6

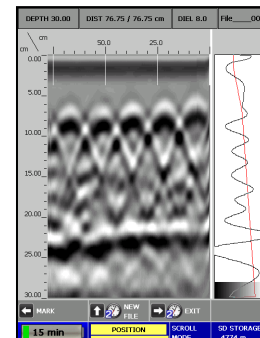
Note: For more detailed information on concrete dielectric values, refer to the GSSI Handbook for “Radar Inspection of Concrete” that was included with your system on the documentation CD.

COLLECT	File_005
PROJECT	Proj_2
DEPTH	40 cm
DIELECTRIC	7.5
AUTO TARGET	ON
DISPLAY	A+B
COLOR	
CHANGE FRACTION	CHANGE VALUE
62 min	12:41:31
	SD STORAGE: 4707 m

CHANGE FRACTION	CHANGE VALUE	QUIT	SET VALUE
62 min	12:41:31		SD STORAGE: 4707 m

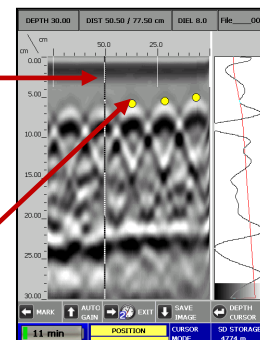
Auto Target: Allowing the Mini HR to Automatically Pick Your Targets

- 1 Highlight Auto Target.
- 2 Press $\leftarrow$ to toggle between ON and OFF.
 - **Off** – No dots and no interpretation will be done by the Mini HR.
 - **On** – The Mini HR will place dots on what it interprets as targets as you collect data, and when pulling the Mini HR straight back.



Backup Cursor

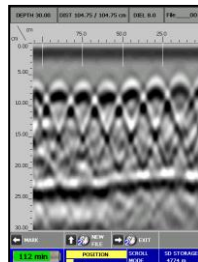
Auto Target Picks



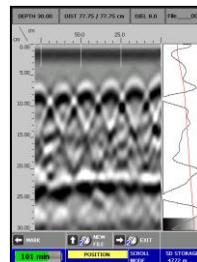
Display: Collecting Data, Display Linescan Data Only or Linescan Data + O-Scope

- 1 Highlight Display.
- 2 Press $\leftarrow$ to toggle between B (data only) and A+B (data + o-scope).

Linescan Data only
display during collection

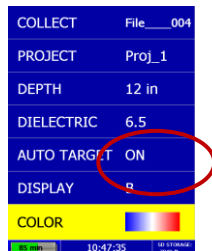


Linescan Data + O-Scope
display during collection



Color: Choosing Data Display Colors

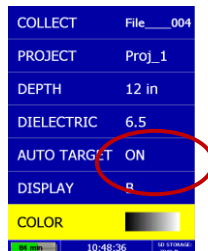
- 1 Highlight Color.
- 2 Press $\leftarrow$ to toggle between five the different color schemes available for data display during collection.



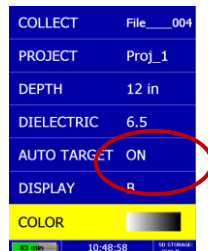
Color table 1



Color table 2



Color table 3



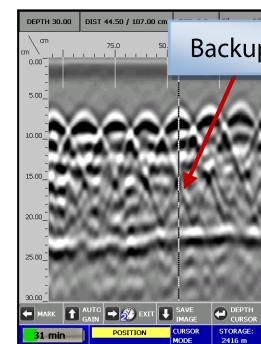
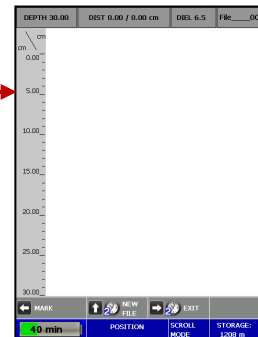
Color table 4



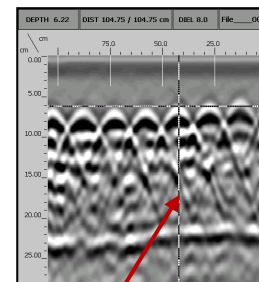
Color table 5

Collecting Data and Marking Your Targets

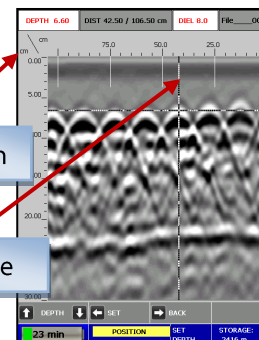
- 1 Highlight Start Collect and press . This screen is displayed.
 - 2 Note the information at the top of the screen:
 - **Depth** – The depth you have chosen for this scan in inches or centimeters.
 - **Dist.** – The distance you have scanned.
 - **Diel.** – The dielectric you have chosen for this scan.
 - **"File ____XXX"** – The filename of this scan if you choose to save this file.
 - 3 Move your cart forward. Data only gets collected when you move the cart forward.
 - 4 Once you start seeing hyperbolas on the screen, pull the system straight back along your survey line. You will see a vertical line (the backup cursor) scroll along your data. When that vertical line is right over the apex of the hyperbola, the center of the antenna is over that target.
- Note:** When you pull the Mini HR straight back, and if you configured Auto Target On, the Mini HR will interpret target locations by placing yellow dots on the targets.
- 5 The center of the antenna is directly under the Mini HR. After marking the location of the target on the ground using the laser (if configured On) at each side of your cart as a guide (or the groove on each side of the cart as a guide if the laser is configured Off). Push the cart straight forward or move the cart to do another line scan. No data will be collected until you have passed the spot where you started to reverse.



- 6** If you wish, you may calibrate depth (optional). Match the target location in the data to its location in your survey area. Drill down to that target and measure the depth.
- a)** Backup the Mini HR and press **↵** “Depth Cursor” to display crosshairs and set them over the target in the data.
 - b)** Press **↑ ↓** to line up the horizontal line in the middle of first dominant color of that target.
 - c)** Press **↵** to Set Depth.
 - d)** Press **↑ ↓** until the Depth in the upper left hand corner matches the measured depth.
 - e)** Press **←** to Set Depth. The vertical scale will adjust and the dielectric will be updated. Please note that this is only an approximation of the true depth because the dielectric constant can change with depth and across the site.
- 7** Press **→** to continue scanning.

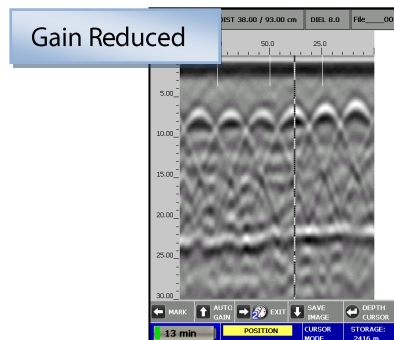
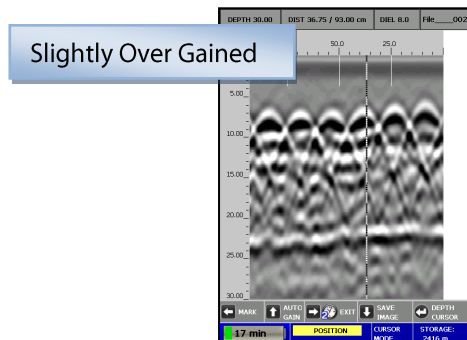


Backup Cursor and
Depth Cursor



8 Auto Gain Feature

- a) At times, while collecting data, the screen may be over or under gained. To correct this, backup the Mini HR and press **↑** Auto Gain. This will correct the gain without changing the dielectric.



- b) Continue forward to collect data with the new gain setting.

9 Save Image

To save an image of the data that was collected, backup the Mini HR and press **↓** Save Image. The file will save as a Bitmap file.

10 Other Options

- While collecting data (with no backup cursor on the screen), press and **HOLD** **↑** (New File) to save your current file. If you have configured Save Prompt On, you will be prompted to confirm the save. Press **↑** to save, **↓** to not save the data. If you have configured Save Prompt Off, it will save automatically.
- Press **←** to place a user mark in your data.
- Press **→** to exit. If you have configured Save Prompt On, you will be prompted to confirm the save. Press **↑** to save, **↓** to not save the data. If you have configured Save Prompt Off, it will save automatically.

Helpful Hint: Make a sketch map of your survey area with immovable objects as reference points. Sketch in your survey files, note the start and stop points, direction of travel, and file name on the map. This will make it much easier to relocate your targets after the survey.

Helpful Hint: The top of a target is generally the middle of the first positive peak for metal targets (rebar, tensioned cable, wires in PVC, conduit). If you are working in a grayscale color table, the positives are white. If the utility is air or gas filled (empty PVC, voids) the top of the target is the middle of the first negative peak (black). Please refer to Appendix C for examples of the different types of data.

Section 4: Data Playback and Review

The Playback menu provides tools to help you interpret your data. None of these functions are required to get interpretable data. They are simply means to help you in the interpretation of the data. These functions do not permanently change your data or affect your saved data in any way. More intensive processing can be done in the RADAN processing program (sold separately).

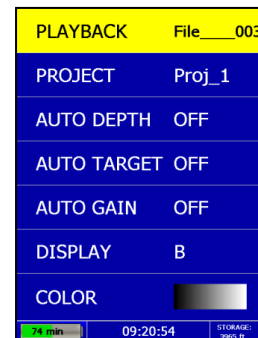
Playback Menu Options

- 1 From the Main Menu, select Playback by pressing →.
- 2 Use ↑ ↓ to highlight the appropriate options under the Playback Menu.

Before you begin to playback data (Playback), you may want to configure the Mini HR to give you some helpful tools in the interpretation of the data.

Project: Selecting a Specific Project to Select a Specific File to Playback

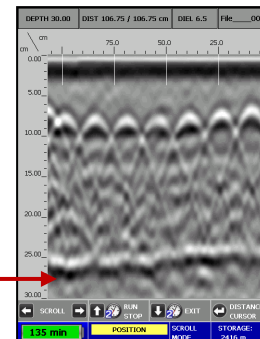
- 1 Highlight Project.
- 2 Highlight and press ← to select the project where the file you want to play back is located.
- 3 Use ↑ ↓ to scroll down or up.
- 4 Highlight and press ← to select the specific file you want to play back.



Auto Depth: Automatically Adjusting Depth Scale

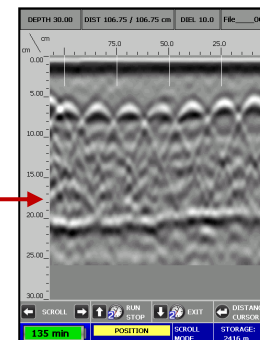
- 1 Highlight Auto Depth.
- 2 Press $\leftarrow$ to toggle between ON and OFF.
 - **OFF** – The Mini HR will use the original value set during collection.

Original Depth Range



- **ON** – The Mini HR will estimate the depth of targets and automatically adjust the Depth Scale.

Depth Range Adjusted



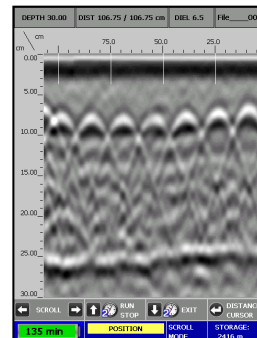
Auto Target: Enabling the Mini HR to Automatically Pick Your Targets

- 1 Highlight Auto Target.
- 2 Press  to toggle between ON and OFF.
 - **OFF** – No dots and no interpretation will be done by the Mini HR.
 - **On** – The Mini HR will place dots on what it interprets as targets.

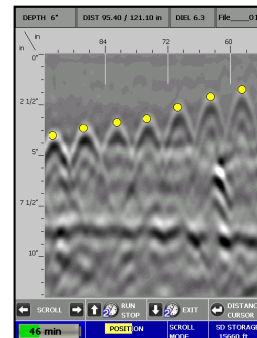
Note: Auto Target is for information purposes only and you should carefully examine your data to ensure that the Mini HR has correctly located objects.

Auto Gain: Enabling the Mini HR to Automatically Configure the Gain

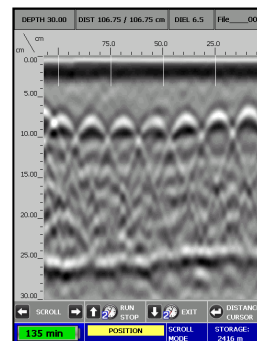
- 1 Highlight Auto Gain.
- 2 Press  to toggle between ON and OFF.
 - **OFF** – The Mini HR will use the gain used during data collection.
 - **On** – The Mini HR will reconfigure the gain.



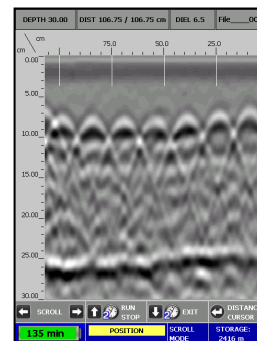
Auto Target Off



Auto Target On



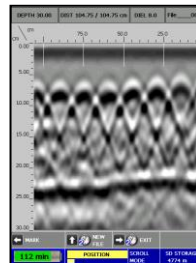
Auto Gain Off



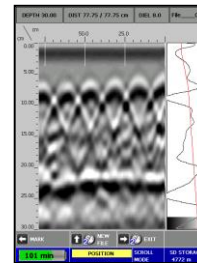
Auto Gain On

Display: Collecting Data, Display Linescan Data Only or Linescan Data + O-Scope

- 1 Highlight Display.
- 2 Press $\leftarrow$ to toggle between B (data only) and A+B (data + o-scope).



Linescan Data only
display during collection



Linescan Data + O-Scope
display during collection

Color: Choosing Data Display Colors

- 1 Highlight Color.
- 2 Press $\leftarrow$ to toggle between five the different color schemes available for data display during collection.



Color table 1



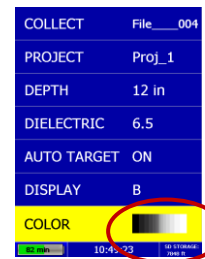
Color table 2



Color table 3



Color table 4



Color table 5

Playing Back the Selected File

- 1 Highlight Playback.
- 2 Press $\leftarrow$ to play back the selected file.

Note at the top of the screen the following information:

- Depth
- Distance
- Dielectric
- File name

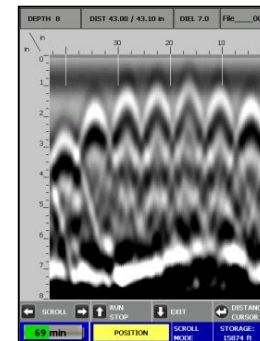
While in Playback, you will have an option to toggle between four different modes by simply pressing the Enter button. The mode you are in will be displayed at the bottom of the screen.

Scroll Mode – Allows you to scroll your data left and right

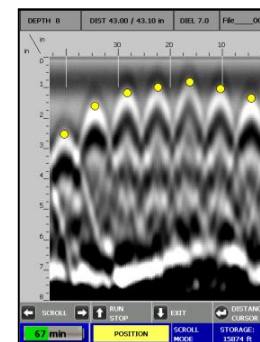
Distance Cursor Mode – Allows you to move a vertical line left and right on your data

Depth Cursor Mode – Allows you to move a horizontal line up and down on your data.

Set Depth Mode – Allows you to set a known depth to your targets.



Data with Auto Target **OFF**



Data with Auto Target **ON**

Scroll Mode

← → – Scroll through your data.

↑ – Run Stop, while your data is scrolling, press Up Arrow to stop and start the scrolling. Hold this button for 2 seconds will load the next file in the project.

↓ – Exit back to the Playback menu. If you added or deleted any targets, the system will prompt you to save your targets. Press ↑ for Yes, ↓ for No.

↶ – Switch to Distance Cursor Mode.



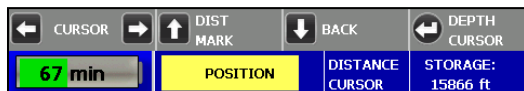
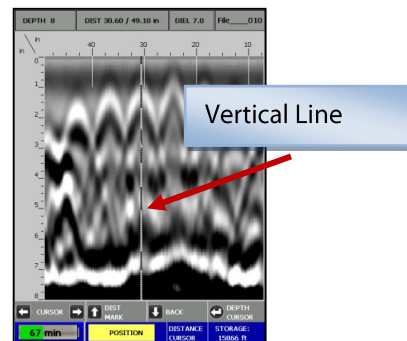
Distance Cursor Mode

← → – Move the vertical line left and right.

↑ – Place a user mark at the location of the vertical line.

↓ – Return to Scroll Mode.

↶ – Switch to Depth Cursor Mode.



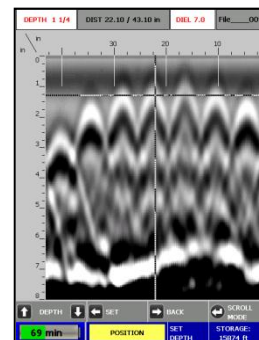
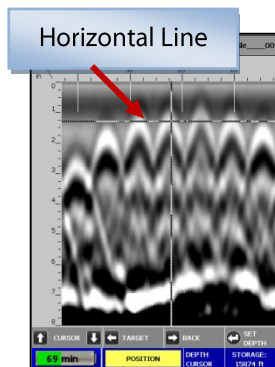
Depth Cursor Mode

- ↑ ↓ – Move the horizontal line up and down.
- ← – Toggles between placing and deleting a dot to mark a target where the cross hairs are.
- – Return to Distance Cursor Mode.
- ↶ – Switch to Set Depth Mode.



Set Depth Mode

- ↑ ↓ – Entering a known depth for the target in the crosshairs. The upper left hand corner displays the depth
- ← – Once the correct depth is displayed, set the depth. Note: If the depth of a target changes, this will automatically change the dielectric.
- – Return to Depth Cursor Mode.
- ↶ – Switch to Scroll Mode.



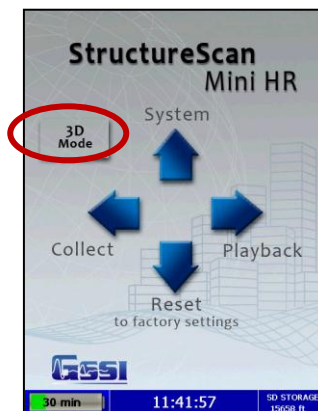
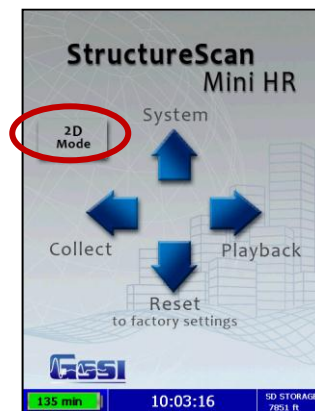
Section 5: StructureScan Mini HR 3D - Collecting and Processing 3D Data

The StructureScan Mini HR can be configured with optional 3D collecting. You may also upgrade to this feature with your current StructureScan Mini HR if you did not originally purchase this feature. Contact your GSSI sales representative for upgrade information.

If you have the 3D feature, there will be a new option on the screen upon booting your StructureScan Mini HR. There will be a button that allows you to toggle between 2D and 3D mode. Simply press the Enter Key ↵ to toggle between 2D and 3D mode.

Configuring the StructureScan Mini HR for 3D Collection

- 1 From the Main Menu, select System by pressing ↑.
- 2 Use ↑ and ↓ buttons to highlight the appropriate options under the System Menu.



Date/Time: Configuring Date and Time

- 1 Highlight Date/Time and press $\leftarrow$.
- 2 Highlight any of the options you wish to change and press $\leftarrow$
The following options will appear at the bottom of the screen:
 - **Change Value** – Use $\uparrow$ and $\downarrow$ to change the value.
 - **Set Value** – Press the Enter button to accept your change.
 - **Quit** – Use $\rightarrow$ to quit **WITHOUT** saving the changes.



- 3 Press $\rightarrow$ to return to the previous menu.

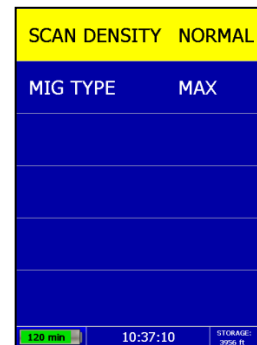
Configuration: Configuring Orientation, Language, Units, Laser, and Sound

- 1 Highlight Configuration and press $\leftarrow$.
- 2 Press $\uparrow$ $\downarrow$ to select the appropriate options:
 - **Orientation** – Press $\leftarrow$ to toggle between Right or Left. This will flip the screen to allow for either right or left handed operation..
 - **Language** – Press $\leftarrow$ to select your language preference.

YEAR	2009
MONTH	AUGUST
DAY	24
HOURL	9
MINUTE	30
TIME ZONE	GMT - 5
160 min	09:30:08
STORAGE: 15881 ft	

ORIENTATION	RIGHT
LANGUAGE	ENGLISH
UNITS	ENGLISH
LASER	ON
SOUND	HIGH
VERSION(s)	1.1.1.5mn
OPTIONS	
120 min	10:36:38
STORAGE: 9956 ft	

- **Units** – Press  to toggle between English or Metric.
 - **Laser** – Press  to toggle laser On or Off. If configured On, the laser will light up in the Collect Menu only. Note: The laser will blink (whether configured On or Off) when battery life is low (10%).
Caution: Do not stare into beam.
 - **Sound** – Press  to toggle between High, Medium, Low, or Off.
 - **Version(s)** – Press  to display the version number of the software. Please visit GSSI's web site for updates to the firmware.
 - **Options** – Press  to configure Scan Density and Migration Type.
 - **Scan Density** – Press  to toggle between Normal and Low.
 - **Normal** – The Mini HR will scan at 8 scans/cm (approximately 240 scans/ft).
 - **Low** – The Mini HR will scan at 4 scans/cm (approximately 120 scans/ft).
 - **Mig Type** – When Migrating the data, whether to use the full samples or half samples.
 - **Max** – Full number of samples (slower processing, but clearer picture).
 - **Half** – Every other sample (faster processing, but not as clear picture).
 - Press  to return to the previous menu.
- 3** Press  to return to the previous menu.



Calibration: Auto Calibrations for Surface and Brightness

- 1 Highlight Calibration and press $\leftarrow$.
 - a) Hold the Mini HR upside down about 1 meter (3 ft) from any surface.
 - b) Press $\leftarrow$. The Auto-Calibration will take several seconds.
- 2 Press $\rightarrow$ to return to the previous menu.

Clear Storage: Deleting Saved Data

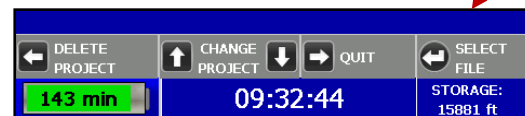
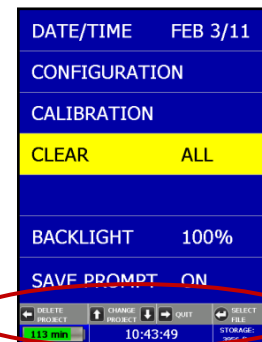
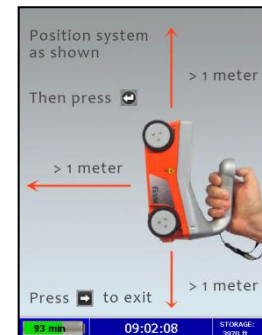
A project is a folder which holds individual files.

- 1 Highlight Clear Storage and press $\leftarrow$.
- 2 At the bottom of the screen, a new set of options is displayed.

Deleting All Projects, Specific Projects or Specific Files

All Projects: If you wish to delete **ALL** of your projects and all the files associated with each project:

- 1 Press $\leftarrow$ when Clear All is displayed.
- 2 Press $\uparrow$ to confirm, $\downarrow$ to cancel the deletion.

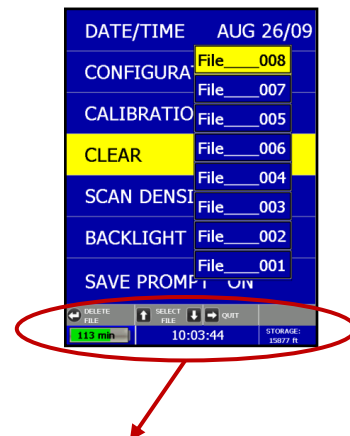


Specific Projects: If you wish to delete a specific project or specific files within a project:

- 1 Press $\uparrow$ $\downarrow$ to select the specific project.
- 2 Press $\leftarrow$ to delete that entire project (and all the files associated with that project).
- 3 Press $\uparrow$ to confirm, $\downarrow$ to cancel.
- 4 At any time, you may press $\rightarrow$ to quit.

Specific Files in a Specific Projects: Deleting specific files within a specific project:

- 1 Press $\uparrow$ $\downarrow$ to select a specific project.
- 2 Press $\leftarrow$ to select that project.
- 3 Press $\uparrow$ $\downarrow$ to select a specific file.
- 4 Press $\rightarrow$ to delete the selected file.
- 5 Press $\uparrow$ to confirm, $\downarrow$ to cancel.
- 6 At any time, you may press $\rightarrow$ to quit.



Backlight: Setting the Brightness of the Mini HR Backlight

- 1 Highlight Backlight.
- 2 Press  to toggle between 25%, 50%, 75%, or 100%.

Save Prompt: Have the System Prompt You to Save or Save Automatically

- 1 Highlight Save Prompt.
- 2 Press  to toggle between ON or OFF.
 - **On** – The system will prompt you to save a file after you choose to save a file.
 - **Off** – The system will save the file automatically after you choose to save a file.

Collecting 3D Data

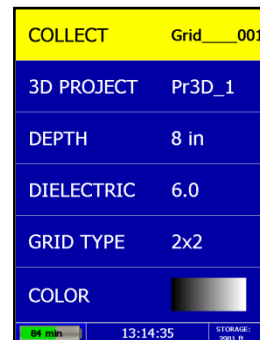
After your configuration is complete, you are now ready to 3D collect data.

Important Note: To assure consistency of StructureScan Mini HR's naming of the files collected and saved, it is highly recommended that the SD card is inserted prior to any data collection.

Collect Menu Options

- 1 From the Main Menu, select Collect by pressing ←.
- 2 Use ↑ ↓ buttons to highlight the appropriate options under the Collect Menu.

Before you begin to collect data (Collect), you may want to configure the Mini HR for a particular project.



3D Project: Saving Files to a Specific Project

- 1 Highlight Project and press ←
- 2 Highlight a Project number and press ← All files that you save during collection will be saved under this project. (Note: you may want to take field notes to document what files went under which project).

Depth: Selecting the Depth Where You are Scanning

- 1 Highlight Depth.
- 2 Press ← to toggle between 8, 12, and 16 inches (20, 30, and 40 cm).

Dielectric: Selecting the Approximate Dielectric of the Area You are Scanning

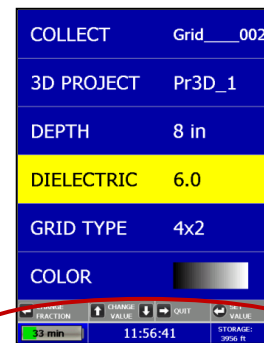
1 Highlight Dielectric and press **↵** Change the Dielectric value by:

- **←** to change the number after the decimal point.
- **↑ ↓** to change the number before the decimal point.
- **→** to quit WITHOUT saving the new value.
- Press **↵** to set your new value.

Having the correct value will improve your depth accuracy. A higher value is necessary young concrete or concrete that is under water.

- Less than 2 months/outside in wet environment, use 9+
- Younger than 12 months/outside, use 7-8
- Older than 12 months/indoor/dry, use 5-6

For more detailed information on concrete dielectric values, refer to the GSSI Handbook for RADAR Inspection of Concrete that came with your system on the documentation CD.



Grid Type: Toggle to Select Grid Size

- 1** Highlight Grid Type.
- 2** Press $\leftarrow$ to toggle between 1 x 1, 2 x 2, or 4 x 2 (30 x 30, 100 x 100, 60 x 60, 120 x 60 cm).

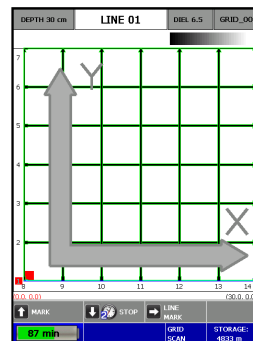
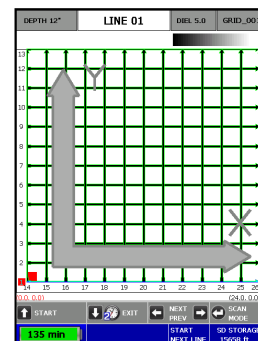
Color: Choosing the Colors You Want Displayed During Data Collection

- 1** Highlight Color.
- 2** Press $\leftarrow$ to toggle between five different color schemes to be displayed during collection.

Collect: Collect 3D Grid

Highlight 3D from the Mini HR start screen and press Collect to begin collecting 3D data.

- 1 Begin by lining up the Mini HR with the appropriate pad.
Using the lasers, line up the laser pointing forward with Line 1.
Line up the side lasers with the start line of the pad.
- 2 Once you have lined up the Mini HR, Press collect and the 3D grid you have chosen will be displayed.
- 3 Press the START  and then roll the Mini HR to the other end of the grid.
The Mini HR will automatically stop collecting once it reaches the end of the grid.
- 4 Return the Mini HR to the start of the grid and line up the lasers with the next line and repeat the process until all the lines needed are collected.



5 Note at the top of the screen. This displays which line number you should be collecting.

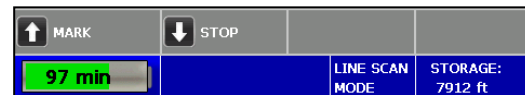
6 You may press the “Next\Prev” buttons, (Left\Right) $\leftarrow \rightarrow$, to scan a specific line. This can be used if you do not collect a line correctly and need to redo a line.

7 Press ↵ Scan Mode, to toggle the view of the scan.

8 During collection, you may:

- Press the Mark Button $\uparrow$ to place a mark in the data as a reference point.
- Press the Stop Button $\downarrow$ and hold for two seconds to stop collecting that particular line because of an obstacle.

9 When you are done collecting all the lines, press and hold the Exit Button $\downarrow$ for two seconds. The Mini HR will save, process, and display the grid (See Playback section for processing saved data).



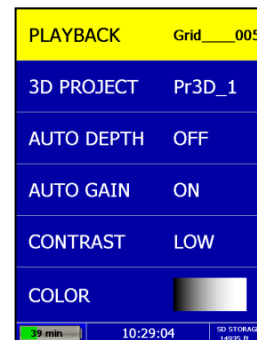
Playback: Processing and Drawing the 3D Grid

There are two ways to process and draw the 3D Grid.

- 1 After you finished collecting the data by pressing and holding the Exit Button  for two seconds. The Mini HR will save, process, and display the grid.

OR

- 2 From the Main Menu, press  to select Playback.
 - a) Highlight 3D Project and press  to select the appropriate project and then select the appropriate Grid.
 - b) Highlight Playback and press  to process and draw the grid.



Other Playback Menu Options

Auto Depth – Press  to toggle between On and Off.

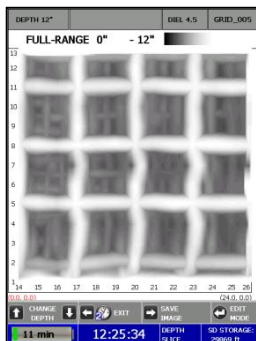
- **On** – The Mini HR will estimate the depth of targets and automatically adjust the Depth Scale.
- **Off** – The Mini HR will use the original value set during collection.

Auto Gain - Press  to toggle between On and Off.

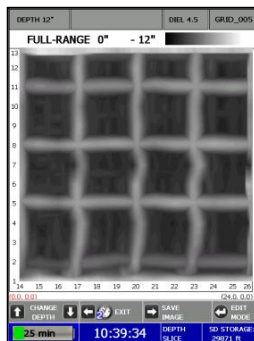
- **On** – The Mini HR will reconfigure the gain.
- **Off** – The Mini HR will use the gain used during data collection.

Contrast - Press  to toggle between Lowest, Low, Medium, High, Highest.

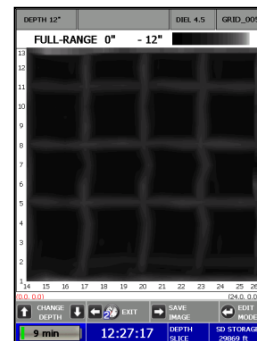
- The lower the contrast, the more “positive” colors are displayed.
- The higher the contrast, the more “negative” colors are displayed.
- See the color scale below to see how much color is displayed as you toggle through these options. Low or Medium are good defaults.



Lowest



Medium



Highest

Color – Press  to toggle between colors for display of the data.

Once the Mini HR has processed the data, it will display the grid. You will be in Depth Slice Mode as noted at the bottom of the screen.

Modes

You may press the Enter button  to toggle between different modes for display and edit purposes. They are: Depth Slice Mode, Cursor Mode, Color Mode, and Select Profile Mode.

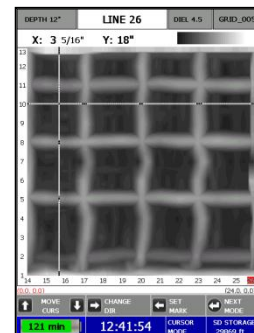
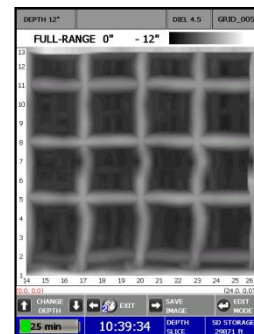
Depth Slice Mode

- 1 The initial display will be the Full Range of the grid. Pressing the  from the Full Range will decrease the range depth by ½ inch (1.27 cm) increments. Pressing the  will slice down through the image in ½ inch (1.27 cm) increments. Note the top of the screen. It will display the depth of the slice or the total depth of the slice.
- 2 Press  and hold for two seconds to Exit.
- 3 Press  to save the image as a .bmp file.
- 4 Press  to toggle to Cursor Mode.

Cursor Mode

This will display crosshairs on the data so that X and Y coordinates located at the top of the screen may be noted as the location of a target.

- 1 Press   buttons to move the crosshair in the Y Direction.
- 2 Press  to change the menu to the X Direction Mode so that you may move the crosshair in the X Directions by pressing  .
- 3 Note the X and Y coordinates at the top of the screen as you move the cross hairs.



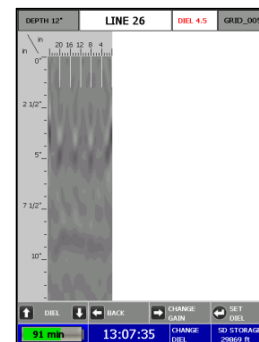
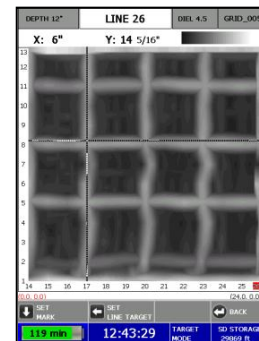
- 4 Press  to set a mark at the location of the crosshairs.
 - Press  to place a dot at the location of the crosshairs.
 - Press  to place a line following the vertical line of the crosshairs.
 - Press  to return back to Edit Mode View.
- 5 Press  to toggle to Color Mode.

Color Mode: Allow to change data color and contrast

- 1 Press  to toggle between different colors.
- 2 Press   to change the contrast of the screen.
- 3 Press  to toggle to Select Profile Mode.

Select Profile Mode: Displays a specific 2D profile on the screen

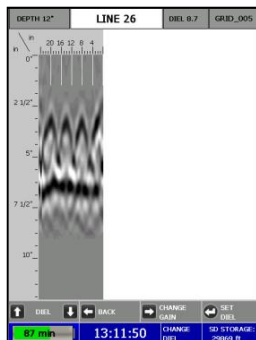
- 1 Press   to select which profile to display.
- 2 Press  to display the 2D profile.
 - Press   to display previous or next 2D Profile.
 - Press  to turn Focus on or off.
 - Press  to return to Select Profile Mode.
 - Press  to change Dielectric.



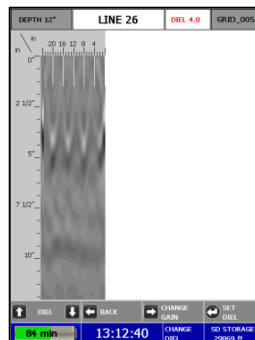
- Press $\uparrow \downarrow$ to change the Dielectric Value.
- Press $\leftarrow$ to return to the previous screen.
- Press $\rightarrow$ to change the gain.
- Press $\swarrow$ to set the new Dielectric.

Press $\swarrow$ to return to Depth Slice Mode

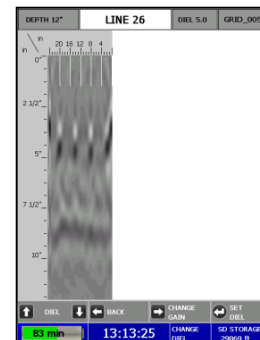
Note: When changing and setting the dielectric, the system will redisplay the data. The goal is to have “dots” instead of hyperbolas’.



Dielectric Too High, Hyperbola tails point down. Adjust Dielectric Down



Dielectric Too Low, Hyperbola tails point up. Adjust Dielectric Up



Dielectric about Right, Displays Dots and no or very little tails.

Section 6: Reset to Factory Settings

This will reset the Mini HR to the factory settings. This will NOT delete the data you have saved.

Simply press the Down Arrow from the Main Menu.

↑ to Confirm.

↓ to Cancel.

Appendix A: Inventory of Your System

- StructureScan Mini HR
- SD card (already inserted into the back of the Mini HR)
- Battery charger (2 slots)
- Battery charger adapter
- International plugs for the battery charger adapter (3)
- Quick Start Guide (this manual)
- USB cable
- Batteries (2)
- GSSI wrist strap
- Training DVD
- Resource CD
- Transit case



Note: If you wish to transfer any saved data to your PC, simply remove the SD card and transfer the data to your PC.

Appendix B: Battery/Charger Information

Your system comes with a two-slotted battery charger and adapter. It also comes with three international interchangeable plugs.

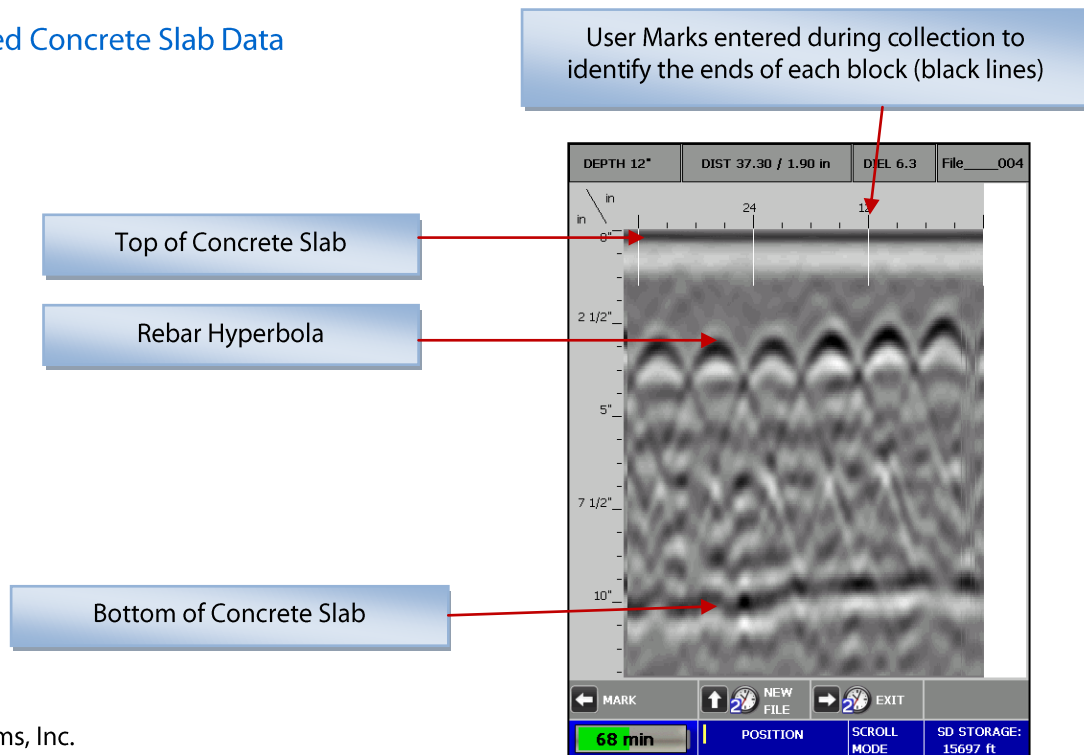
- To charge your batteries, simply attach the adapter to the charger and plug in the adapter. Insert one or both batteries. The batteries will charge one at a time. The bars located at the top of the batteries will indicate how much charge is left in the battery.
- At times, you may wish to re-calibrate a battery. Simply insert a battery into the slot on the left and press the RECAL button
- **LED Lights** – As the batteries are charging, the LED lights for each slot will do different things.
 - No Lights – No Batteries
 - Green Flash – Fast Charging
 - Green Solid – Fully Charged
 - Yellow Flash – Recalibrating
 - Yellow/Green – Recalibrated
 - Yellow Solid – Standby
 - Red Flash – Error
- If the charger remains plugged in and there are NO batteries in the charger, the charger will eventually begin to “beep.” Since this can be annoying, it is recommend that you unplug the charger when not in use.



Appendix C: Sample Data

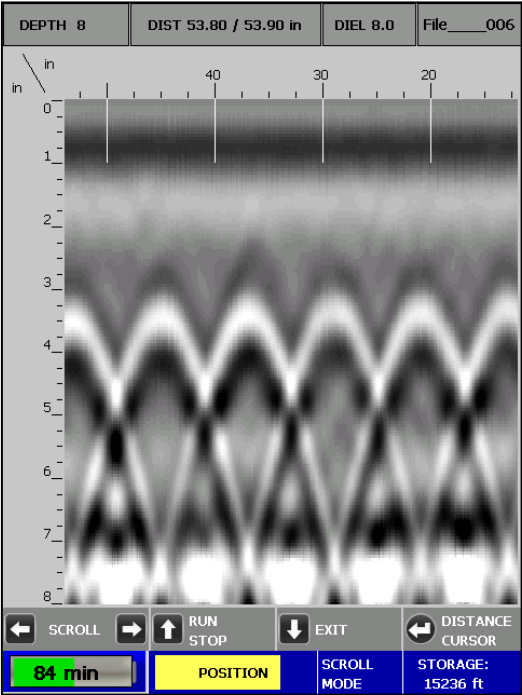
These examples are presented for informational use only. Conditions may be different at your site that may cause the images to look different from the data shown here.

Example 1 – Reinforced Concrete Slab Data



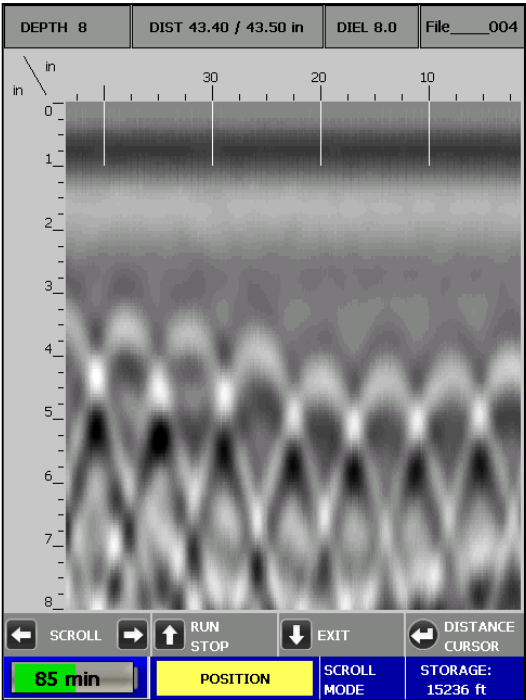
Example 2 – Rebar Data

This is an example of rebar. Note the distance between the peaks here (as well as at the location if you marked the area) is approximately 8 inches. Also note the approximate depth, the brightness of the hyperbolas, and they tend to be at a constant depth over short distances.



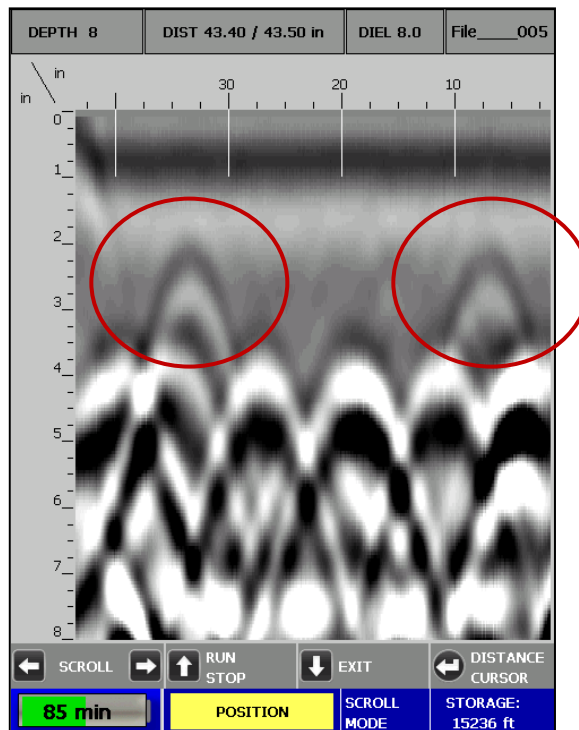
Example 3 – Mesh Data

This is an example of mesh. Note the distance between the peaks here (as well as at the location if you marked the area) is approximately 6 inches. Note the approximate depth and the brightness of the hyperbolas. Wire mesh depths may be more uneven than rebar.



Example 4 – PVC

This is an example of two PVC pipes. Note that they tend to be dimmer than rebar.



Appendix D: Laser Information

Laser Information: General

- Do not intentionally point the laser into your eye. Do not look directly into the laser light source.
- Do not place the laser in a position that may cause anyone to stare into the laser beam intentionally or unintentionally.
- Always power down the unit when not in use.
- Never view a laser through an optical device such as binoculars or a microscope.

Laser Information: Specific for the StructureScan Mini HR

- Do not remove or deface any product labels.
- This product contains lasers used as guides. Do not stare into any of the beams.
- You may configure the lasers on or off. From the Main menu:
 - 1 Select **↑** Configure
 - 2 Press **↑ ↓** and highlight Configuration and press Enter.
 - 3 Press **↑ ↓** and highlight LASER
 - 4 Press **↵** to toggle the laser On or Off.

**WARNING!**

Read and understand all instructions. Failure to follow all instructions may result in serious personal injury.

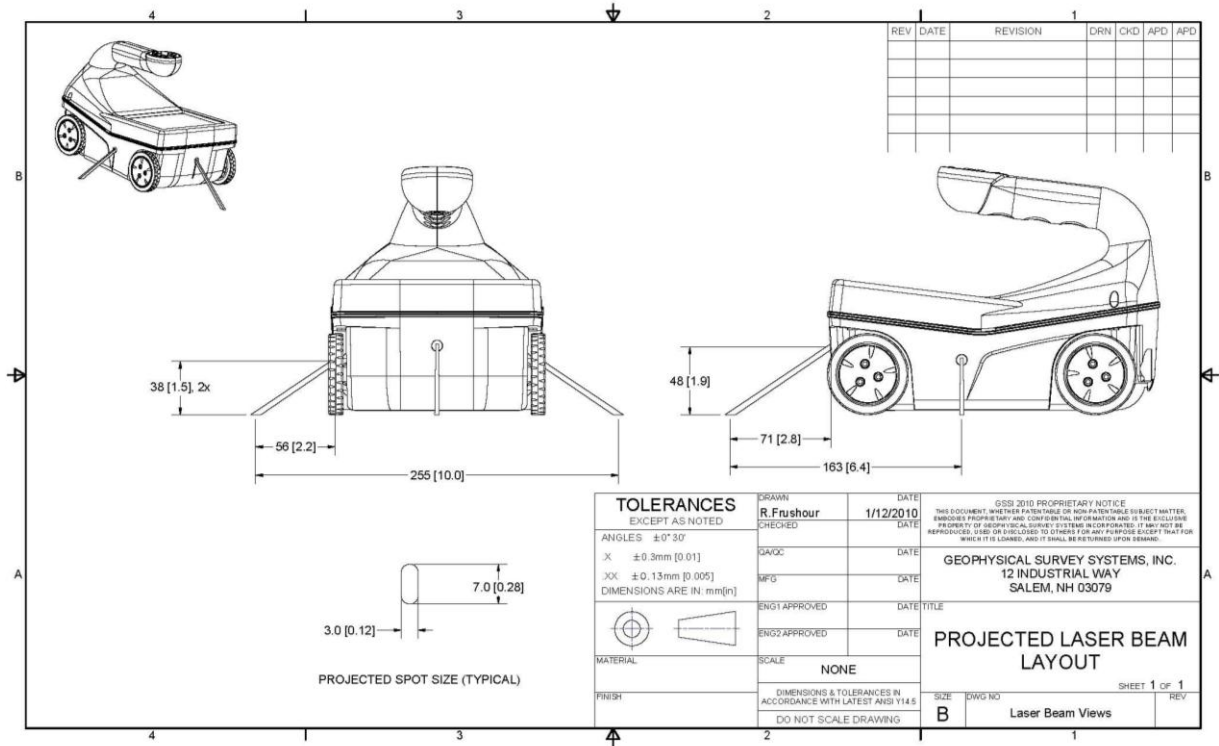
**CAUTION!**

Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

LASER RADIATION**DO NOT STARE INTO THE BEAMS****CLASS 2 LASER PRODUCT**

- This product is designed to be opened by Field Service Professionals employed and/or trained by Geophysical Survey Systems, Inc. only. Do not open this product. Tampering with this product voids all warranties.
- The laser has no serviceable parts.
- The laser will flash several times during system boot up.
- The laser will blink (whether Laser is configured On or Off) when the battery life is low (10%).
- Remove the battery whenever transporting the product.
- Remove the battery whenever doing any self maintenance to the product, such as cleaning the windows of the lasers.
- If the lasers are configured On and the lasers are not working during collection of data, do not open the product. The product must be serviced by an authorized GSSI employee or representative only.
- If the lasers are configured On, they will work only while in the Collect Menu option.
- The lasers will remain on (if Laser is configured On) even if the unit goes into sleep mode.

Laser Information: Projected Laser Beam Layout



Laser Information: Label Placement Diagrams

